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INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N° 8– Réseaux I

Préparé par : Brian ANTOINE

Présenté à: M. Ismaël SAINT AMOUR

Niveau : L3

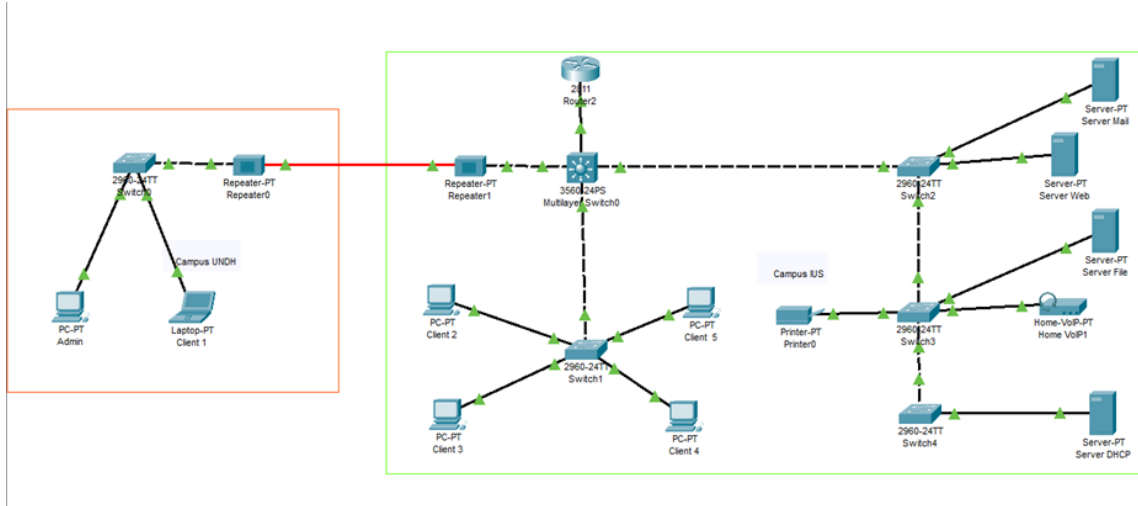
Date: 04/01/2026

L'objectif de ce TD est de :

1. Créer et gérer des VLAN sur des switches
2. Comprendre le rôle des VLAN pour segmenter un réseau.
3. Apprendre à créer des VLAN et à affecter des ports aux VLAN correspondants.
4. Configurer le routage inter-VLAN sur un routeur (Router-on-a-Stick)
5. Comprendre comment un routeur peut permettre aux VLAN de communiquer entre eux.
6. Configurer des sous-interfaces avec encapsulation 802.1Q.
7. Configurer les trunks entre switches et vers le routeur
8. Apprendre à utiliser les trunks pour transporter plusieurs VLAN sur une seule liaison physique.
9. Comprendre le concept de VLAN natif et VLAN autorisés sur les trunks.
10. Attribuer des adresses IP et des gateways aux hôtes
11. Configurer les PCs dans différents VLAN avec des IP correctes.
12. Vérifier la communication intra-VLAN (même VLAN) et inter-VLAN (différents VLAN)

1. Reproduction de la topologie en créant et configurant plusieurs VLAN sur les switches. Assignons les ports aux VLAN correspondants, puis testons et gérons la connectivité entre eux à l'aide d'un routeur.

❖ Reproduction de la topologie :



➤ Tests de vérifications

■ Vérification des Vlan :



Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-R2>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-R2>show vlan
```

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	1000001	1500	-	-	-	-	-	0	0



Switch2

Physical Config CLI Attributes

IOS Command Line Interface

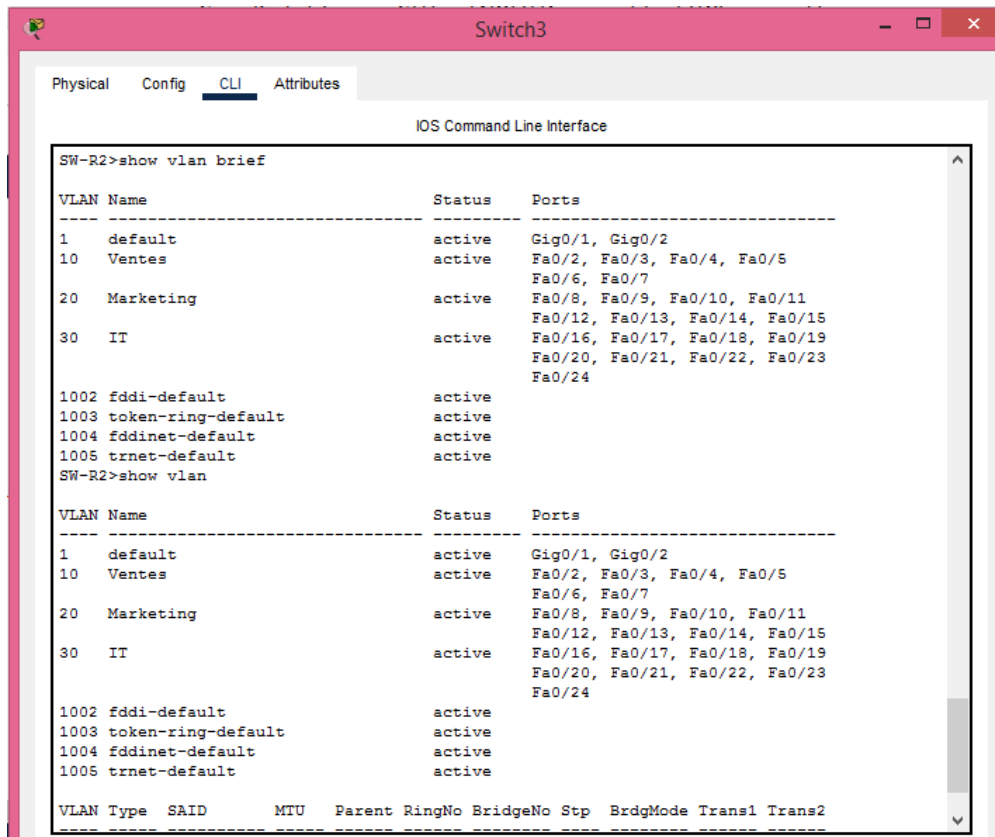
```
SW-R2>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

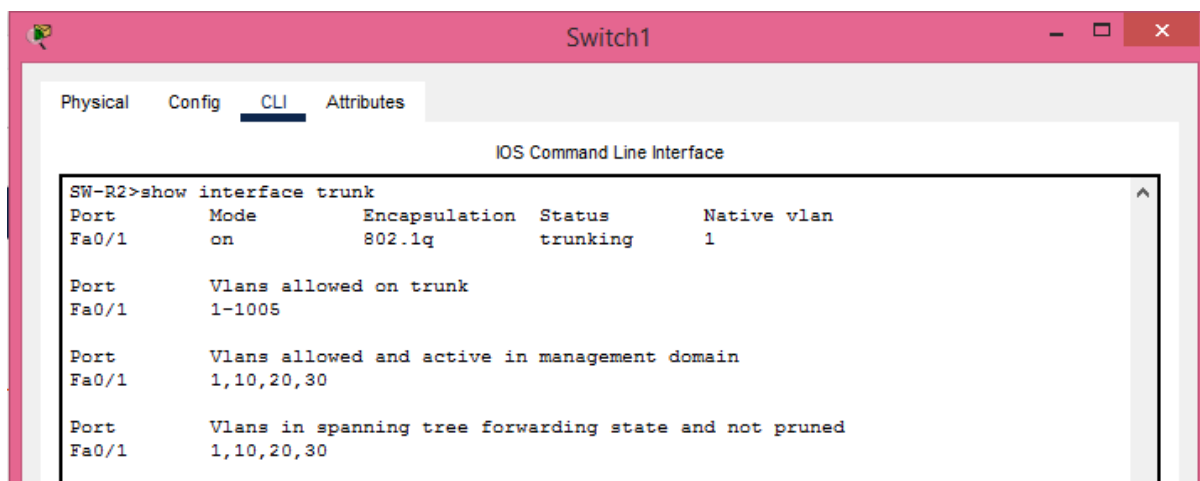
```
SW-R2>show vlan
```

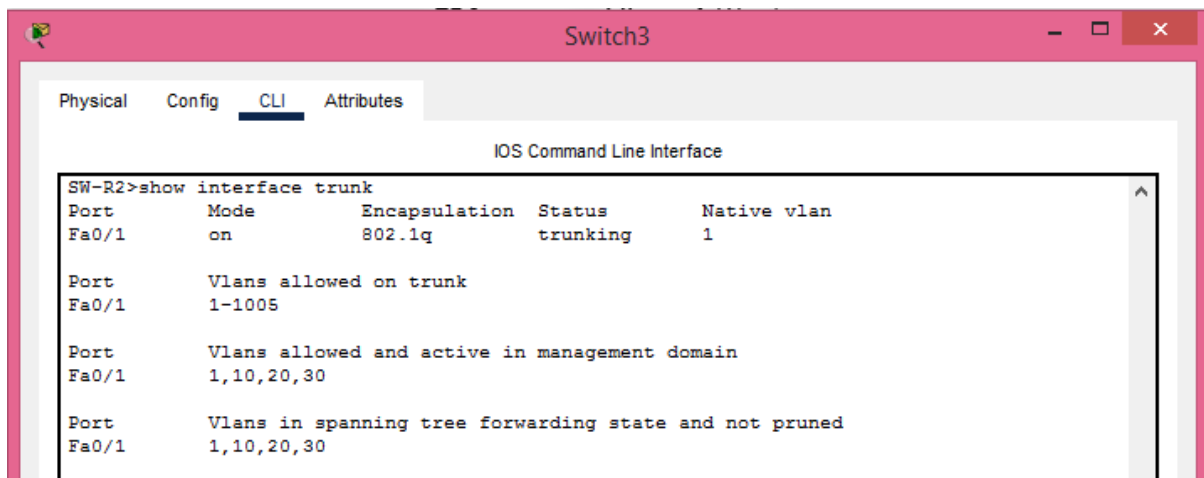
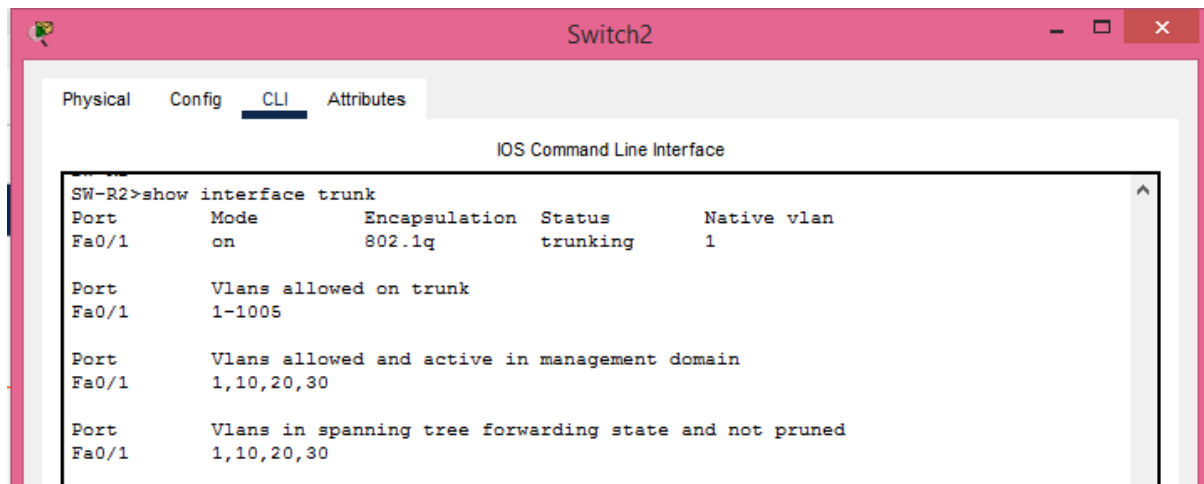
VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	Ventes	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7
20	Marketing	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15
30	IT	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	1000001	1500	-	-	-	-	-	0	0

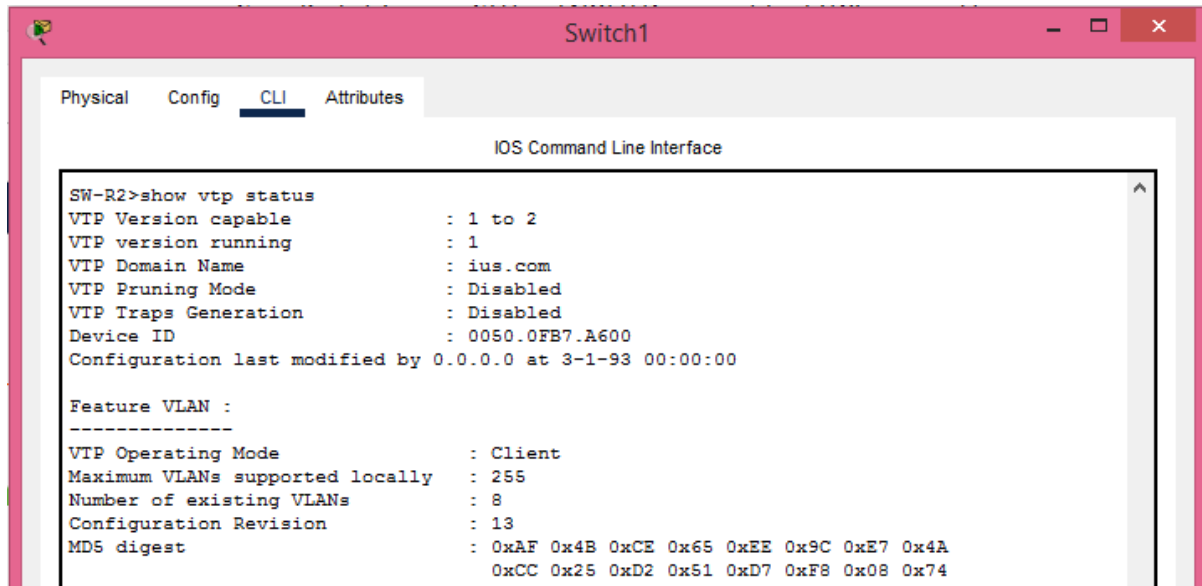


- Vérification des trunks :





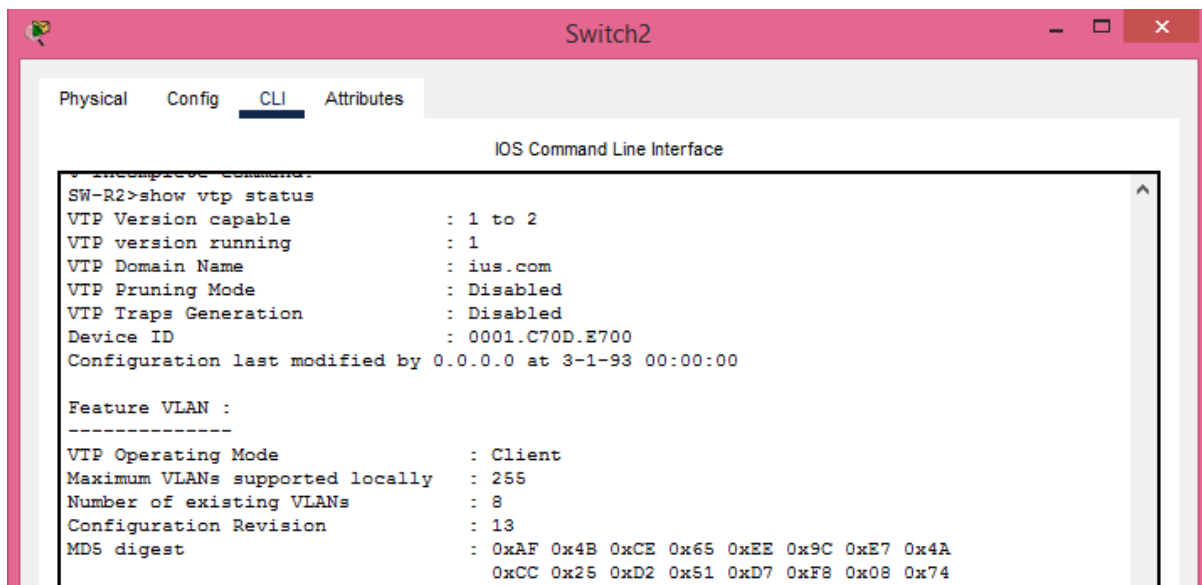
■ Vérification VTP :



The screenshot shows the CLI of Switch1. The 'CLI' tab is selected. The command 'show vtp status' has been entered, displaying the following information:

```
SW-R2>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0050.0FB7.A600
Configuration last modified by 0.0.0.0 at 3-1-93 00:00:00

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision    : 13
MD5 digest               : 0xAF 0x4B 0xCE 0x65 0xEE 0x9C 0xE7 0x4A
                        : 0xCC 0x25 0xD2 0x51 0xD7 0xF8 0x08 0x74
```



The screenshot shows the CLI of Switch2. The 'CLI' tab is selected. The command 'show vtp status' has been entered, displaying the following information:

```
SW-R2>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.C70D.E700
Configuration last modified by 0.0.0.0 at 3-1-93 00:00:00

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision    : 13
MD5 digest               : 0xAF 0x4B 0xCE 0x65 0xEE 0x9C 0xE7 0x4A
                        : 0xCC 0x25 0xD2 0x51 0xD7 0xF8 0x08 0x74
```

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-R2>show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ius.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0002.4ABC.3200
Configuration last modified by 0.0.0.0 at 3-1-93 00:13:29

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 8
Configuration Revision   : 0
MDS digest               : 0x6B 0x43 0xE6 0x23 0xB2 0xC6 0xD2 0x0B
                        : 0x5C 0x0F 0x11 0x3A 0xE7 0xF3 0x5D 0xE9
```

■ Vérification STP :

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
4 vlans          0      0      0      8      8

SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short

Name              Blocking Listening Learning Forwarding STP Active
-----
VLAN0001          0          0          0          1          1
VLAN0010          0          0          0          1          1
VLAN0020          0          0          0          5          5
VLAN0030          0          0          0          1          1
-----
4 vlans          0          0          0          8          8
```


Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	1	1
VLAN0010	0	0	0	2	2
VLAN0020	0	0	0	1	1
VLAN0030	0	0	0	3	3
4 vlans	0	0	0	7	7

Switch3

Physical Config CLI Attributes

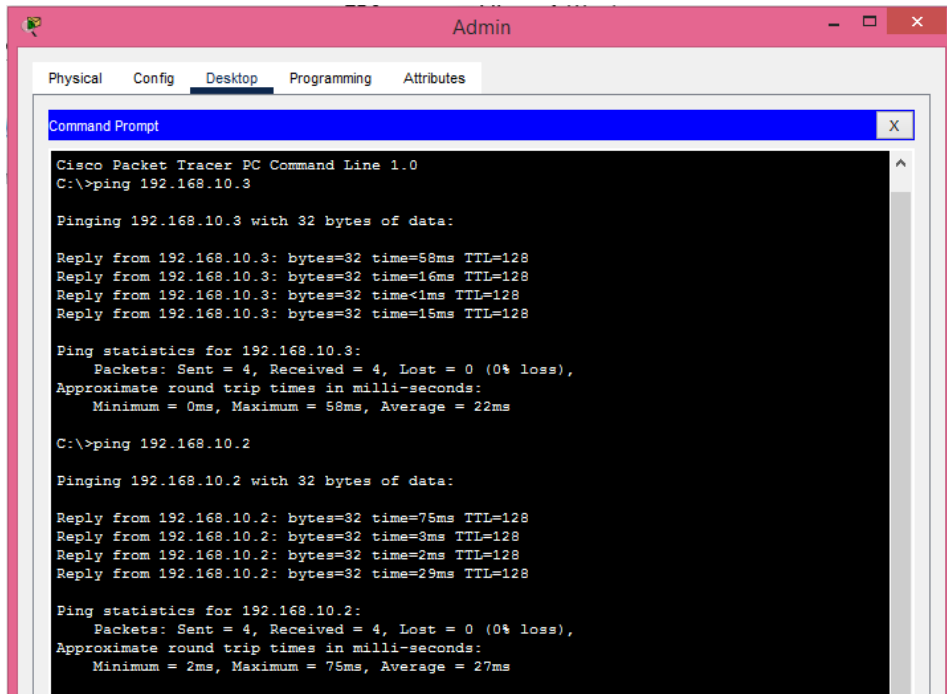
IOS Command Line Interface

```
SW-R2>show spanning-tree summary
Switch is in pvst mode
Root bridge for: Marketing IT
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	1	1
VLAN0010	0	0	0	2	2
VLAN0020	0	0	0	1	1
VLAN0030	0	0	0	4	4
4 vlans	0	0	0	8	8

❖ Testons la connectivité

✓ Test de connectivite sur Vlan 10 :



The screenshot shows the 'Admin' window of Cisco Packet Tracer. The 'Desktop' tab is selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=58ms TTL=128
Reply from 192.168.10.3: bytes=32 time=16ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 58ms, Average = 22ms

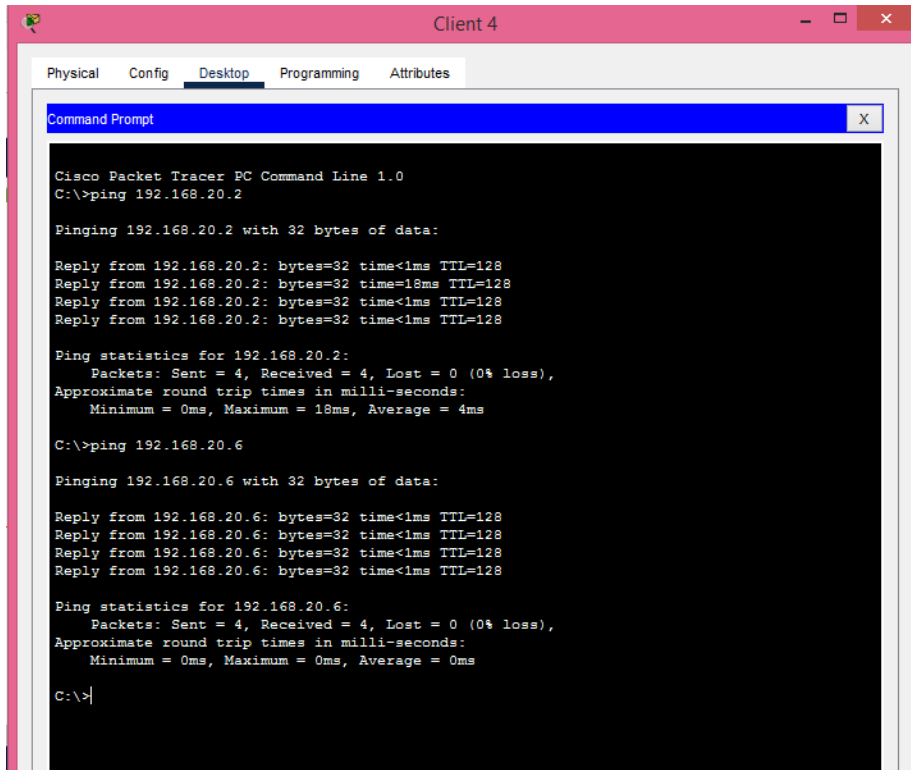
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=75ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time=29ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 75ms, Average = 27ms
```

✓ Test de connectivite sur Vlan 20 :



The screenshot shows the 'Client 4' window of Cisco Packet Tracer. The 'Desktop' tab is selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=18ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 4ms

C:\>ping 192.168.20.6

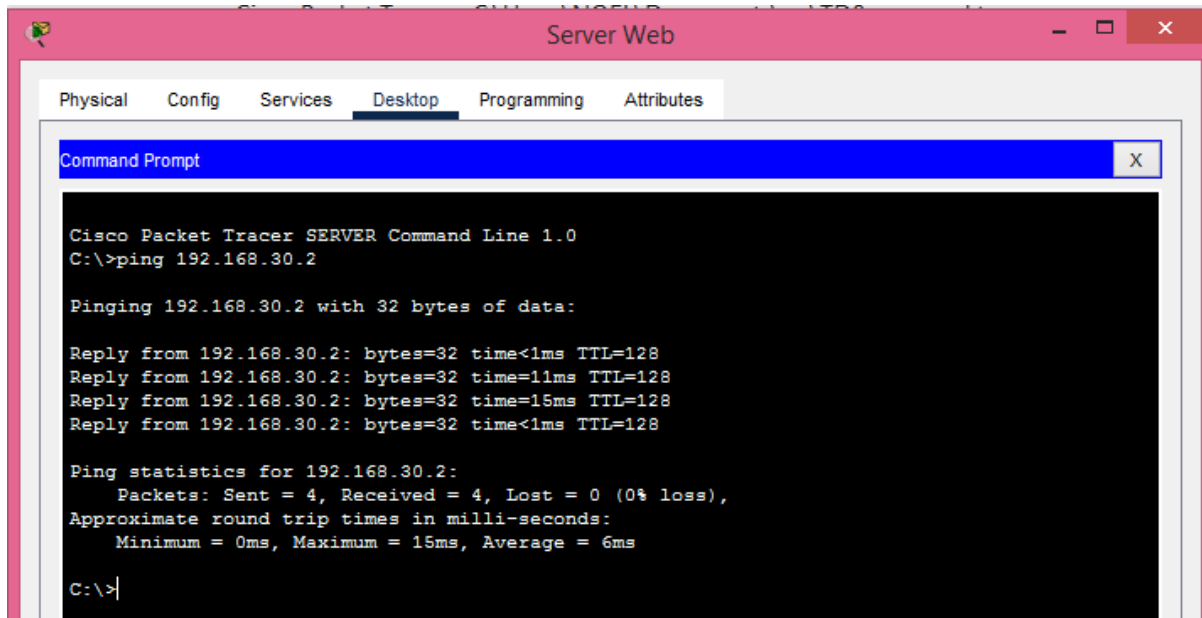
Pinging 192.168.20.6 with 32 bytes of data:

Reply from 192.168.20.6: bytes=32 time<1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128
Reply from 192.168.20.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

✓ Test de connectivite sur Vlan 30 :



The screenshot shows a 'Server Web' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command to 192.168.30.2, resulting in four successful replies with 0% loss.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time=11ms TTL=128
Reply from 192.168.30.2: bytes=32 time=15ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 6ms

C:\>
```

Conclusion

Ce TD m'a permis de passer de la simple connexion de câbles à la véritable gestion d'une architecture. J'ai également appris à **créer des VLAN**, affecter des ports, configurer des liaisons trunks pour transporter plusieurs VLAN, et assurer le routage inter-VLAN grâce à un routeur. En manipulant les VLAN, j'ai réalisé à quel point la segmentation est cruciale pour qu'une entreprise fonctionne bien.